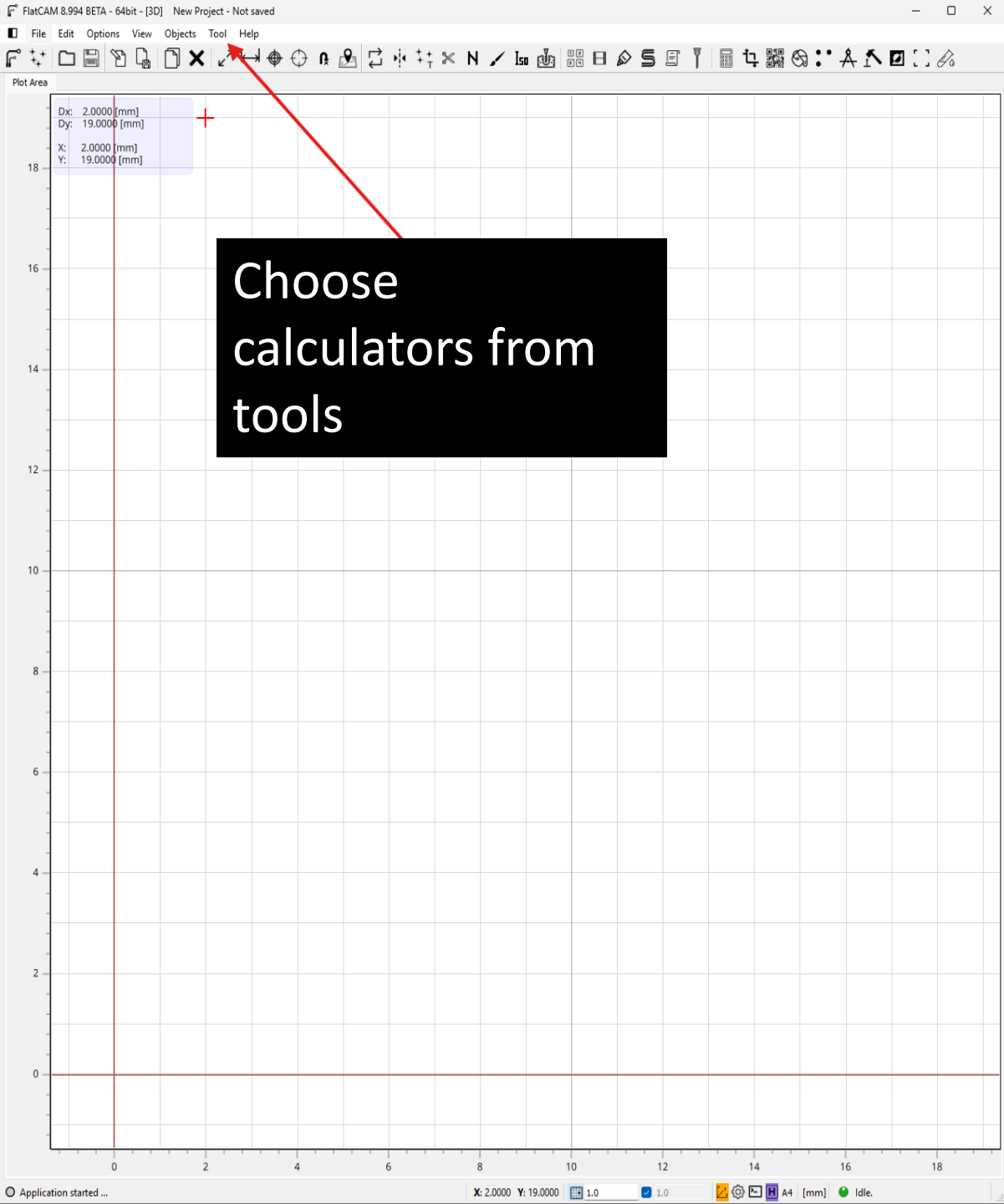


Short info from Flatcam 8.994 Beta

- **Open up Flatcam on the computer**
- **Important to have good file names for the different files.**
- **Gerber:** Typically define copper layers in a circuit board.
- **Excellon:** (drill file): Contain drill specifications, size and coordinates.
- **G-Code:** CNC machine instructions for cutting and/or drilling.

1.1 Calculate tool Diameter



Calculators

Units Calculator

MM

INCH

0.0000

0.0000

V-Shape Tool Calculator

Tip Diameter: 0,2000

Tip Angle: 30

Cut Z: 0,0500

Tool Diameter: 0,0000

Calculate

ElectroPlating Calculator

Area Calculation:

☒ Dimensions

☐ Area

Board Length: 10,0000 cm

Board Width: 10,0000 cm

Current Density: 13,0000 ASF

Copper Growth: 10,0000 um

Current Value: 0,0000 A

Time: 0,0000 min

Calculate

22.5 Tip diameter:
0.2mm

20 Tip Angle: 60

Cut Z:0.5

17.5 Make sure to note
down the given
Tool diameter.

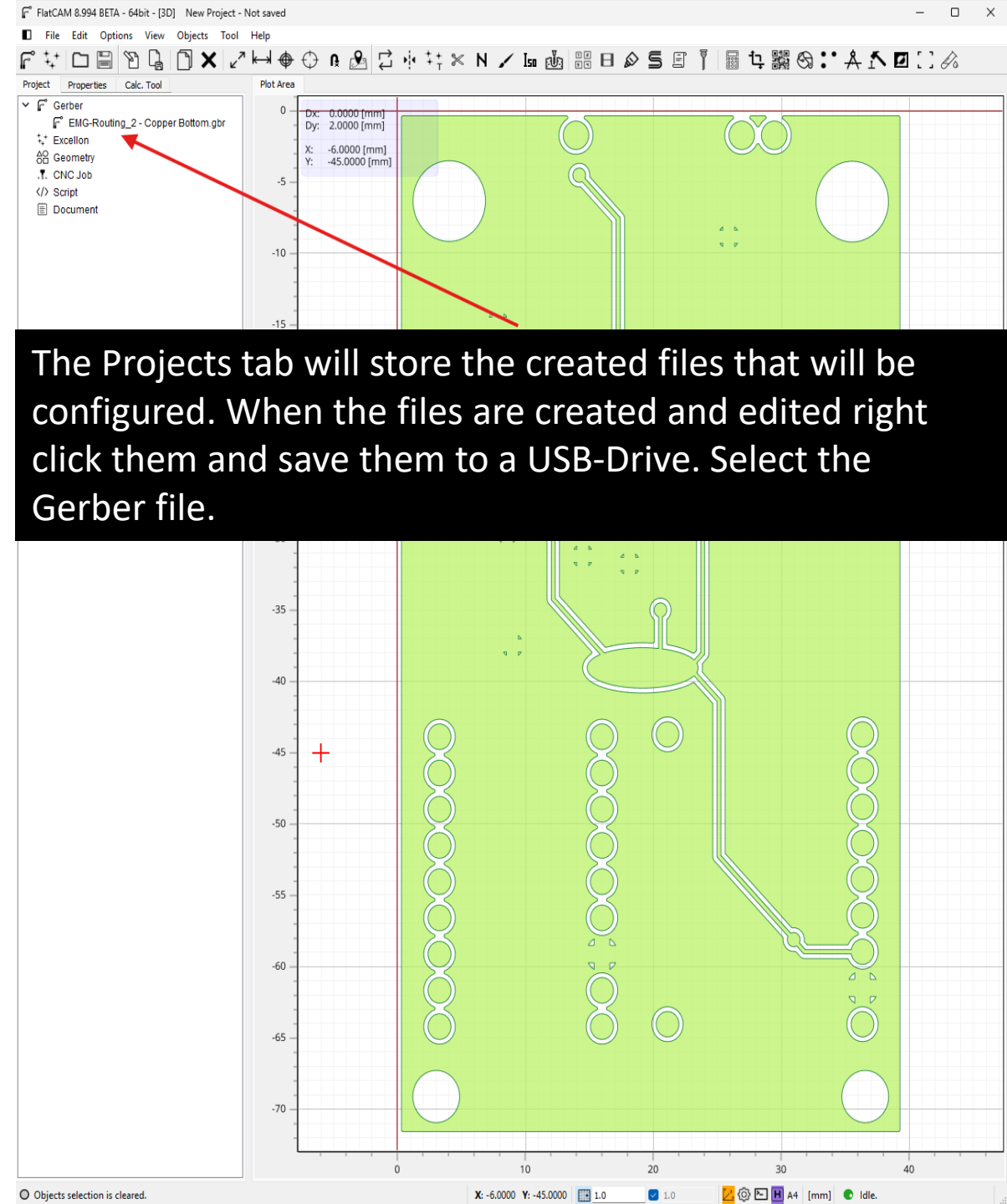
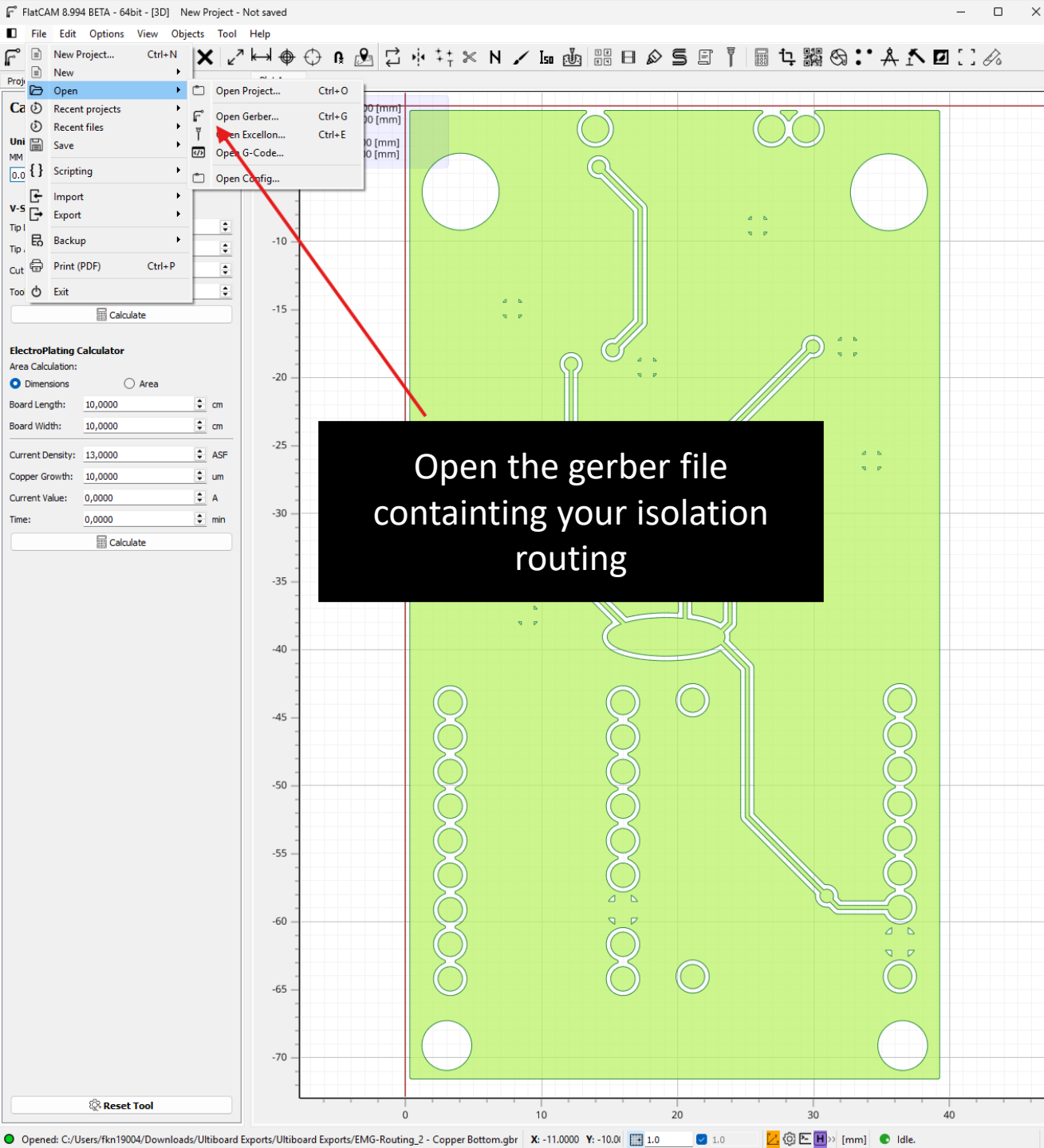
15

12.5

10

1.2 Isolation Routing

- This is where the the traces will be created for the circuit board.
- Isolation routing is the operation of cutting copper around traces for electrical isolation.





Gerber Object

Basic

Plot Options: ☒ Solid

☐ Multi-Color

Name: EMG-Routing_2 - Copper Bottom.gbr

Plot: ☒

Gerber Editor

PROPERTIES

TOOLS

Isolation Routing

NCC Tool

Cutout Tool

UTILITIES

Transformations

1.0000

Scale

(0.0, 0.0)

Offset

Transformations

This will open the gerber object.

Under Tools select isolation routing to Open up the isolation Tool

Used the earlier calculated Tool diameter and input the value given in

Tool Dia:Calculated value

Select Generate geometry. This will create a geometry object in the project tab. Will open automatically.

Isolation Tool

Basic

GERBER:

EMG-Routing_2 - Copper Bottom.gbr

Tools Table

#	Diameter	TT
1	0.1	V

Tool order: ☐ No ☐ Forward ☒ Reverse

Add from DB

Tool Dia: 0,1000

Optimal

Search and Add

Pick from DB

Delete

Parameters for: Tool 1

Passes: 1

Overlap: 10,000 %

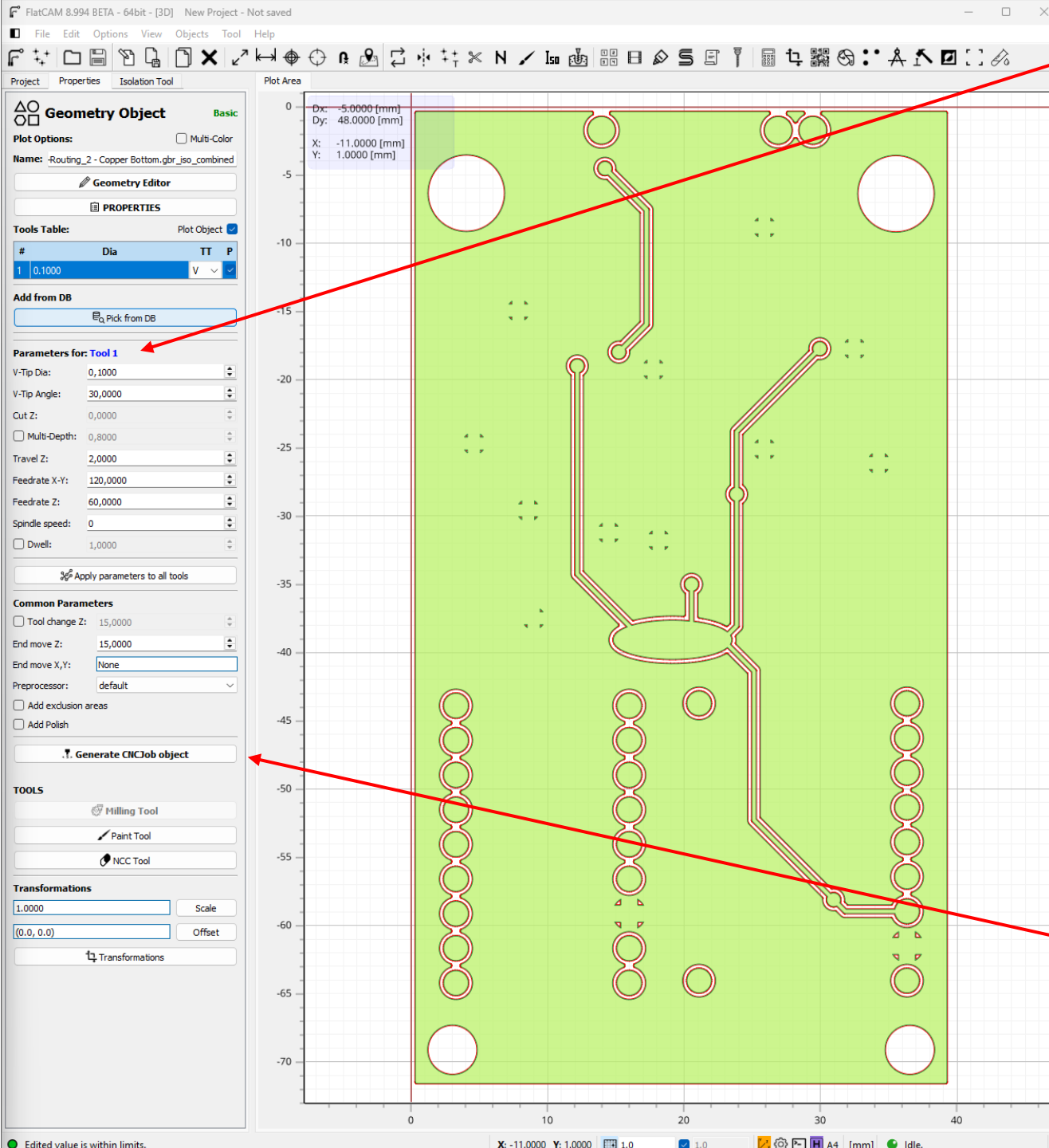
Apply parameters to all tools

Common Parameters

☒ Combine

☐ Check validity

Generate Geometry



Under Parameters for:Tool 1

Fill the following values:

V-Tip Dia:0.02

V-tip angle:60

Feedrate X-Y: 170

Feedrate Z: 170

Spindle speed: 11000

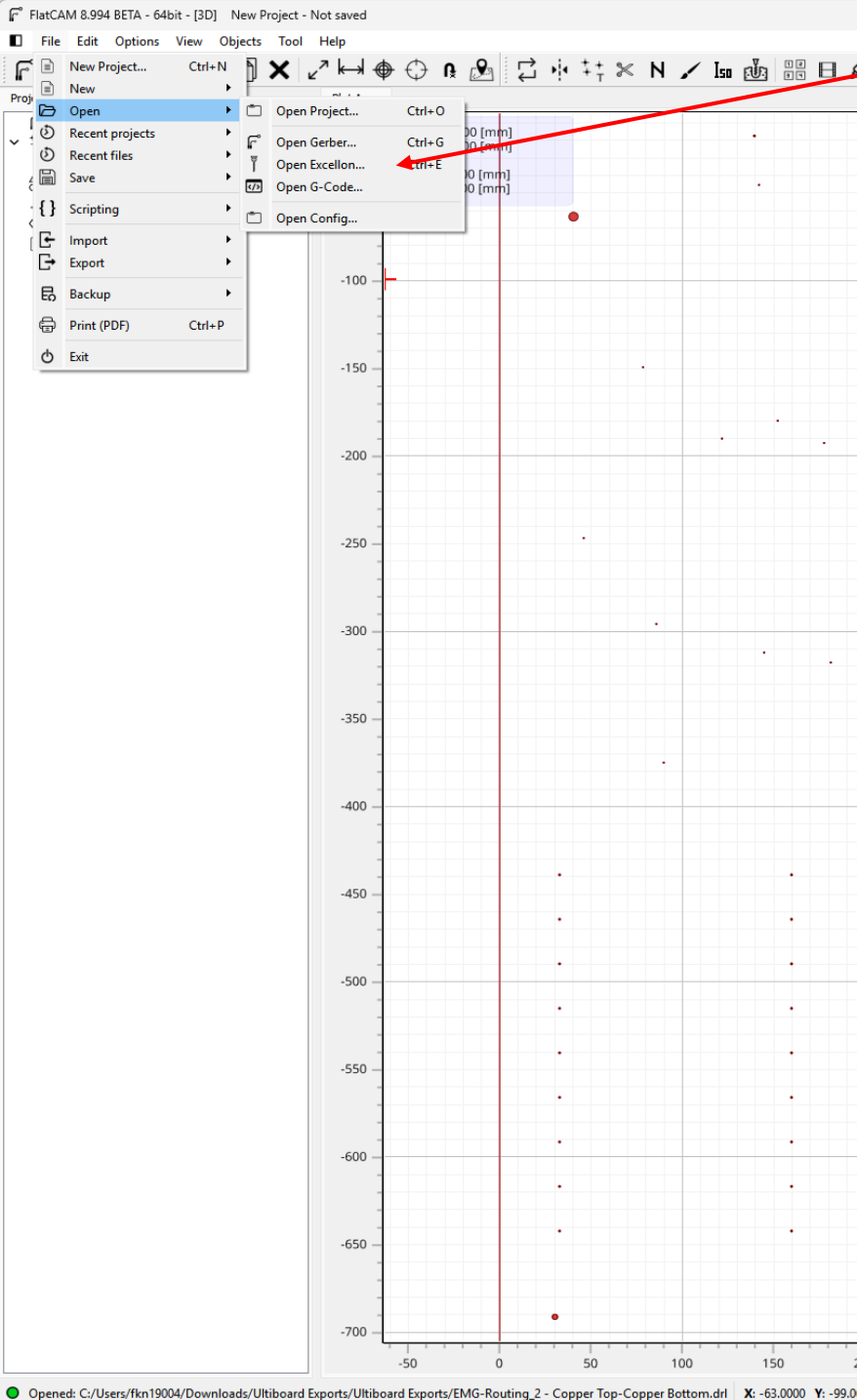
Select: Generate CNCJob object

This will Add the CNC Job to the Project tab

CNC job

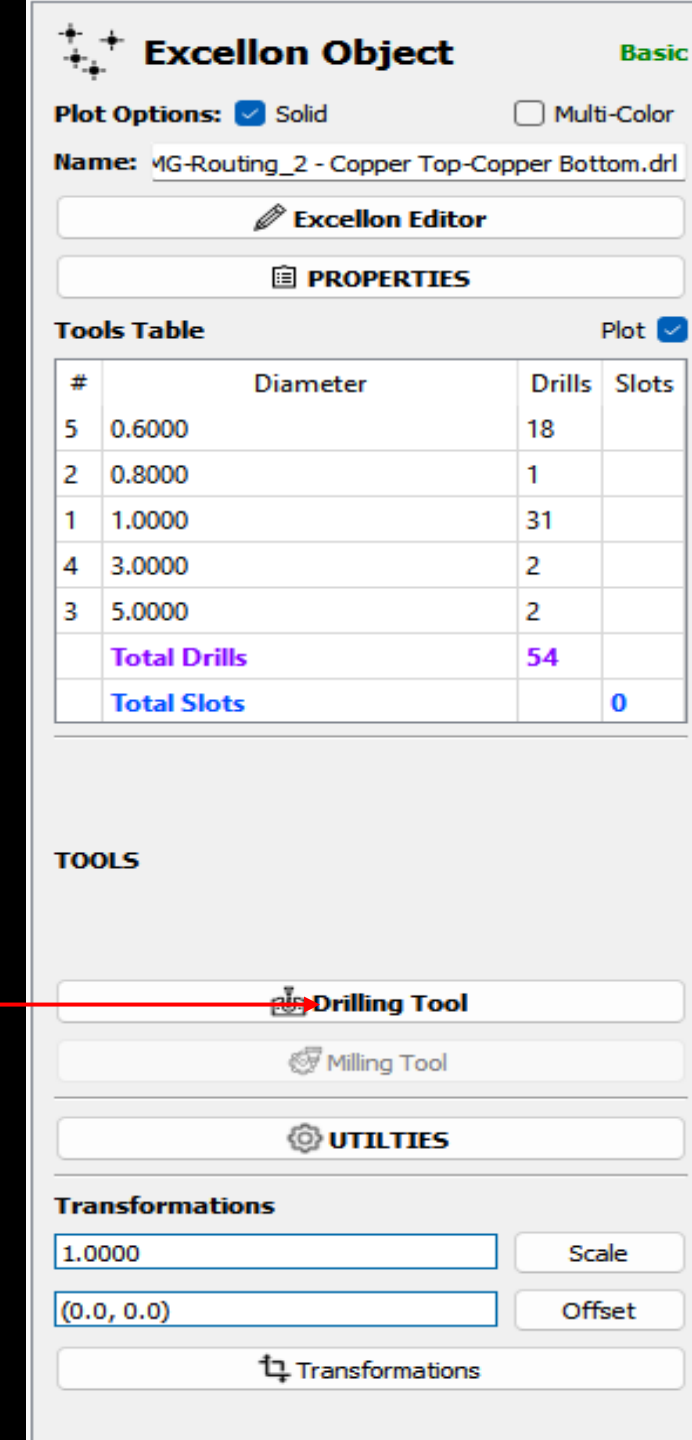
- CNC Job Objects are a simple representation of G-Code in FlatCAM.

1.3 Drilling



Open Excellon to create the drilling project and then select it from project. This will open up the Excellon object.

Select the Drilling Tool



Excellon Object Basic

Plot Options: ☒ Solid ☐ Multi-Color

Name: MG-Routing_2 - Copper Top-Copper Bottom.drl

Excellon Editor

PROPERTIES

Tools Table Plot ☒

#	Diameter	Drills	Slots
5	0.6000	18	
2	0.8000	1	
1	1.0000	31	
4	3.0000	2	
3	5.0000	2	
Total Drills		54	
Total Slots			0

TOOLS

Drilling Tool

Milling Tool

UTILITIES

Transformations

1.0000 Scale

(0.0, 0.0) Offset

Transformations

To change or add drilling diamaters select the Excellen editor

To add a drill select the Tool dia and change to the drill size that is used and then select add

Select exit editor to get back to the Excellon object

Excellon Editor

Name: MG-Routing_2 - Copper Top-Copper Bottom.drl

Tools Table

#	Diameter	D	S
1	0.6000	18	
2	0.8000	1	
3	1.0000	31	
7	10.0000	0	
4	2.0000	0	
5	3.0000	2	
6	5.0000	2	
Total Drills		54	
Total Slots			0

Convert Slots

Add/Delete Tool

Tool Dia: 10.0000 **+ Add**

Delete Tool

Exit Editor

Select Drilling Tool

Excellon Object Basic

Plot Options: ☒ Solid ☐ Multi-Color

Name: MG-Routing_2 - Copper Top-Copper Bottom.drl

Excellon Editor

PROPERTIES

Tools Table Plot ☒

#	Diameter	Drills	Slots
5	0.6000	18	
2	0.8000	1	
1	1.0000	31	
4	3.0000	2	
3	5.0000	2	
Total Drills		54	
Total Slots			0

TOOLS

Drilling Tool

Milling Tool

UTILITIES

Transformations

1.0000 Scale

(0.0, 0.0) Offset

Transformations

Drilling Tool Basic

EXCELLON:
EMG-Routing_2 - Copper Top-Copper Bottom.dl

#	Diameter	Drills	Slots
1	1.0	31	
2	0.8	1	
3	5.0	2	
4	3.0	2	
5	0.6	18	
Total Drills		54	
Total Slots			0

Tool order: ☒ No ☐ Forward ☐ Reverse

Search DB

Parameters for: **Multiple Tools**

Cut Z: -1,7000

☐ Multi-Depth: 0,7000

Travel Z: 2,0000

Feedrate Z: 300,0000

Spindle speed: 0

☐ Dwell: 1,0000

Offset Z: 0,0000

Apply parameters to all tools

Common Parameters

☐ Tool change

Tool change Z: 15,0000

End move Z: 0,5000

End move X,Y: None

Preprocessor: default

☐ Add exclusion areas

. Generate CNCJob object

Reset Tool

Cut Z: to the same value as the thickness of the copper laminate (must be negative value)

Travel Z:2mm

Feedrate Z:170 recommended

Spindle speed: 11000

Select Generate CNCJob object

Board Cutout 1.4



Gerber Object

Basic

Plot Options: ☒ Solid ☐ Multi-Color

Name:

Plot: ☒

 **Gerber Editor**

 **PROPERTIES**

TOOLS

 **Isolation Routing**

 **NCC Tool**

 **Cutout Tool**

 **UTILITIES**

Transformations

 **Transformations**

Select Cutout Tool

This will open the Cutout PCB

Cutout PCB

Source Object:

Kind: ☒ Single ☐ Panel

Type: ☒ Gerber ☐ Geometry

 EMG-Routing_2 - Copper Bottom.gbr

Convex Shape: ☐

Cutout Tool

Tool Dia:

 Search and Add

 Pick from DB

Tool Parameters:

Cut Z:

☒ Multi-Depth:

Margin:

Gap size:

Gap type: ☒ Bridge ☐ Thin ☐ M-Bites

Automatic Bridge Gaps:

Gaps:

 **Generate Geometry**

☐ **Generate Geometry**

Manual Bridge Gaps:

Big cursor: ☒

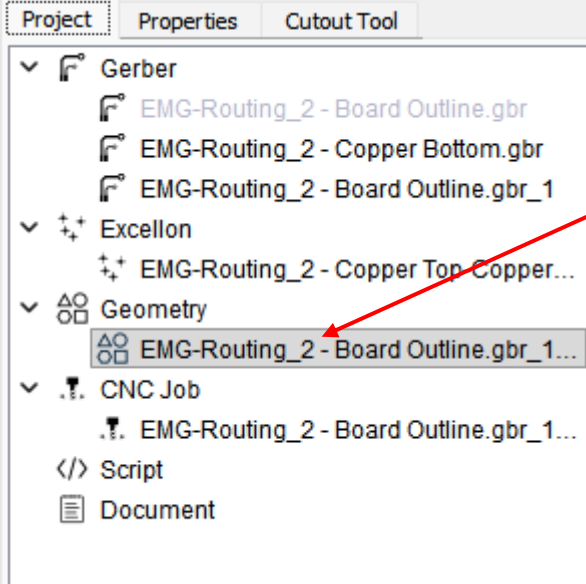
☐ **Generate Manual Geometry**

 **Manual Add Bridge Gaps**

Select the Size of the cutout tool by changing the Tool dia value.

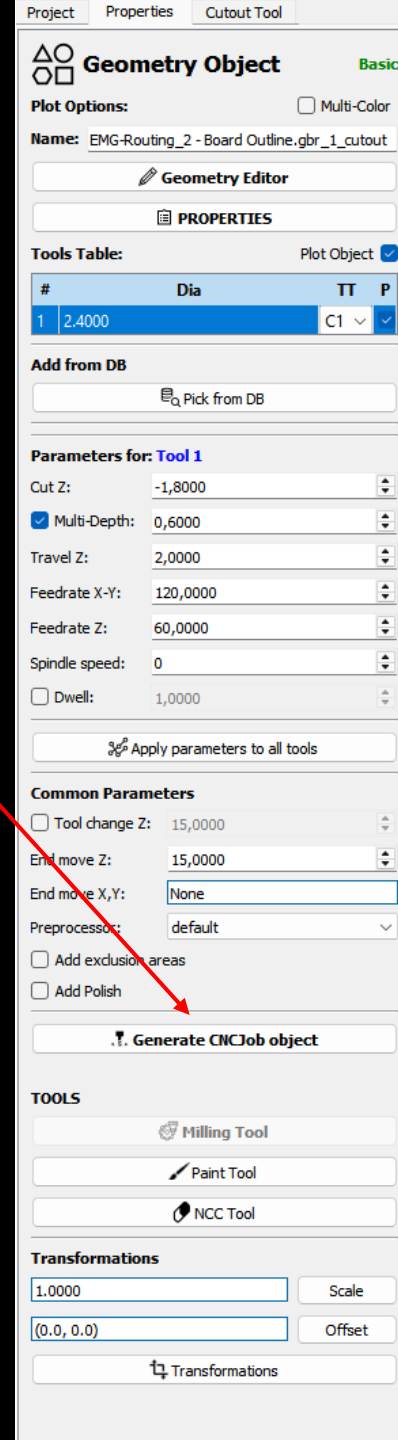
Specify a gapsize. 2 times the dia of the cutting tool is a good value

Specify the amount of gaps along the edge. 2 (top and bottom), 2 (left and right) or 4, one on each side. Select generate geometry. This will add the Board outline to the project files as a geometry project.



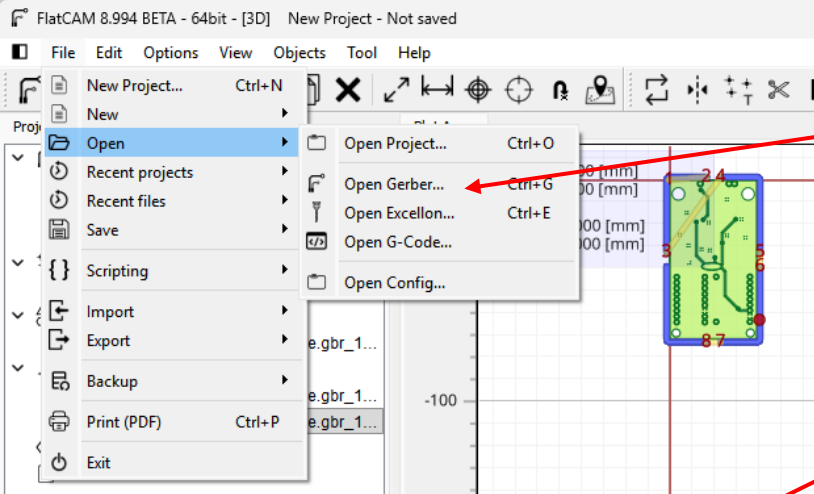
Select the board outline geometry file.

Generate the CNCJob object and save it to the USB-Drive



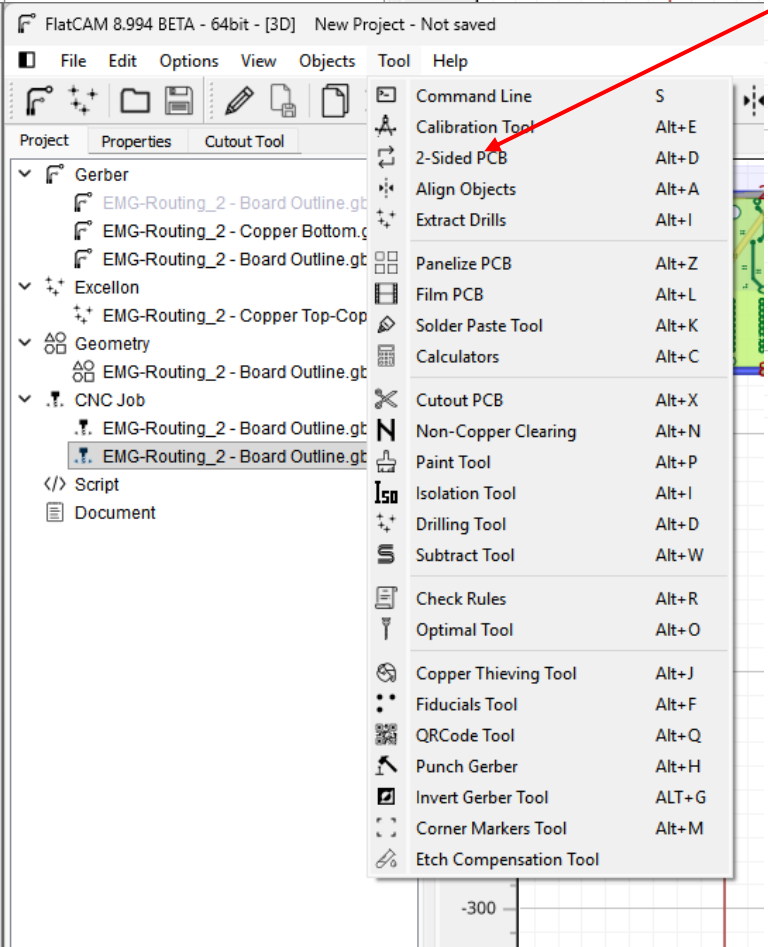
1.5 2-Sided PCB

- Stop here if PCB is 1 sided and continue to Wegster operation

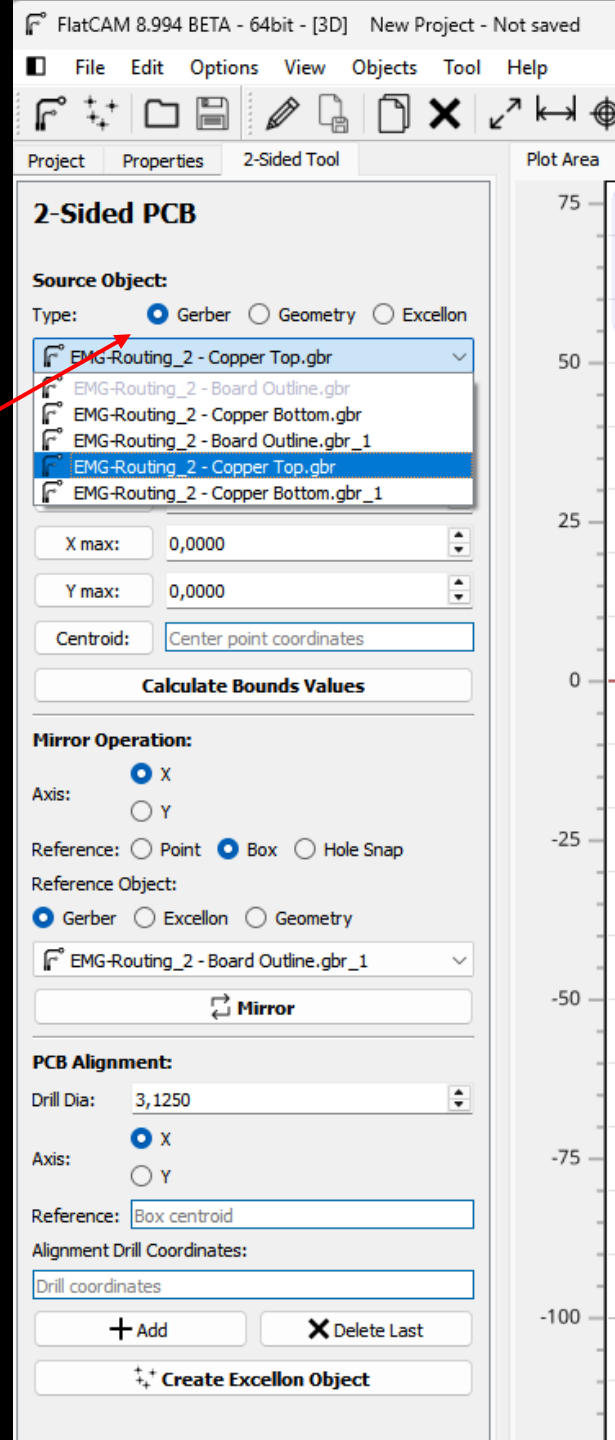


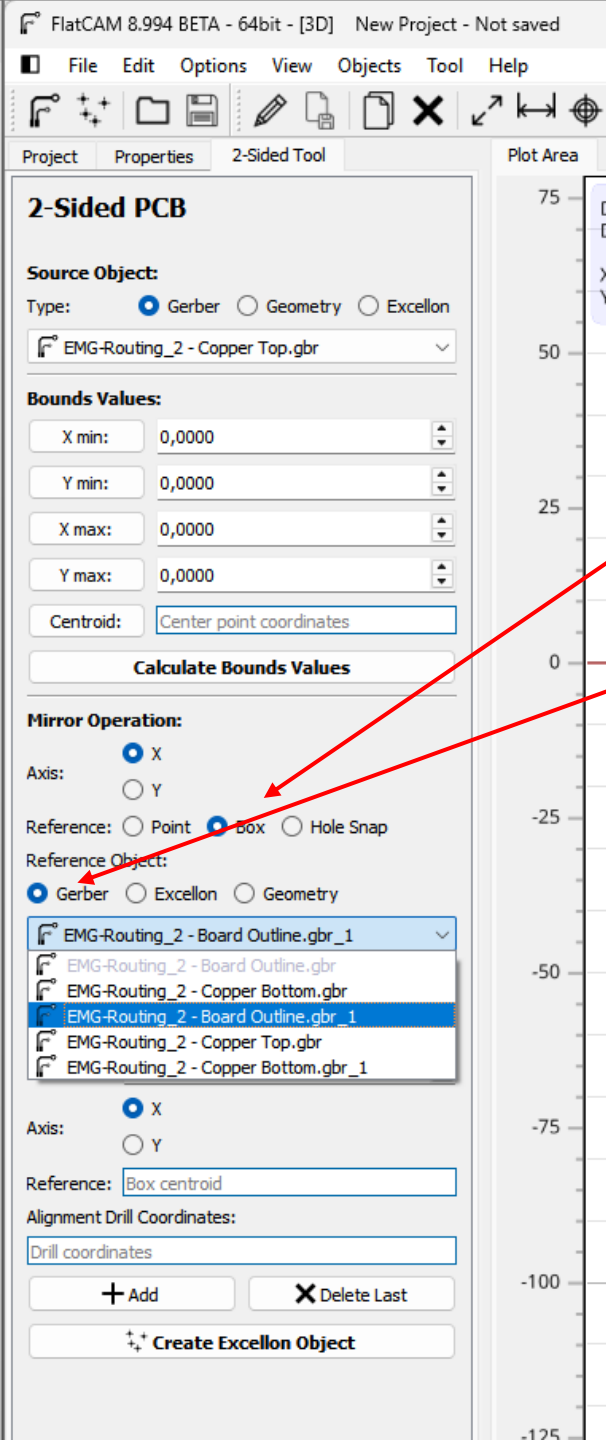
For 2 sided PCB select the Gerber file with the other side of the PCB

Then from Tool select the 2-Sided PCB option.



Select Gerber under Type
And pick the side that you are going to work on. If you started with the bottom, pick the top here.





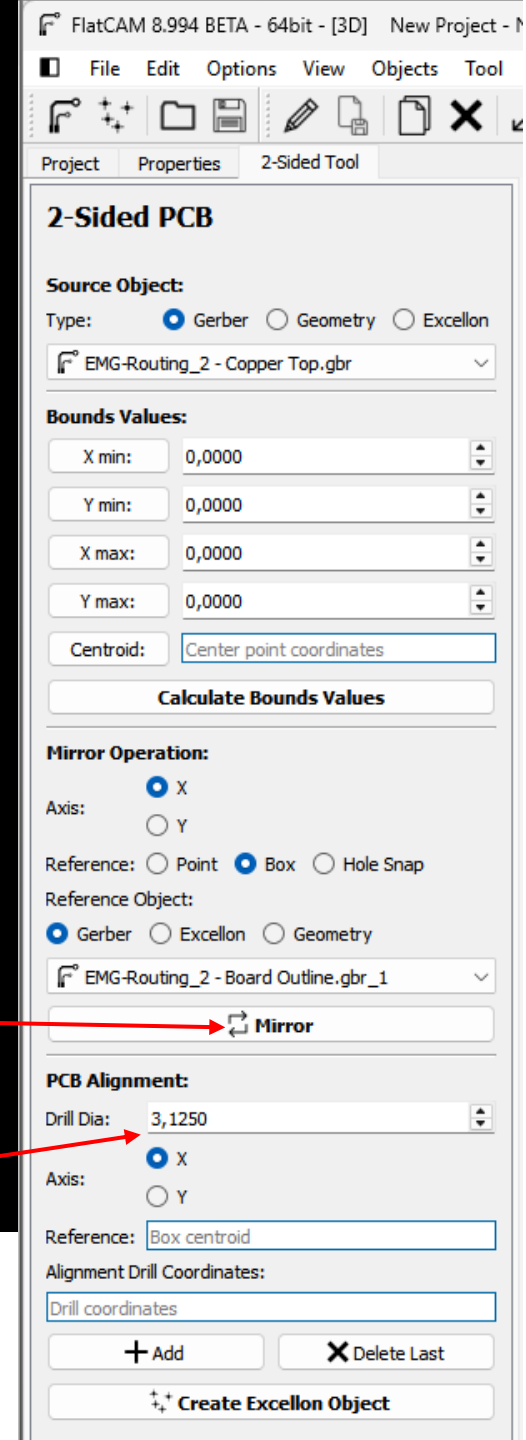
Select box at reference

Select gerber under reference object:

Select the Board outline

Select Mirror

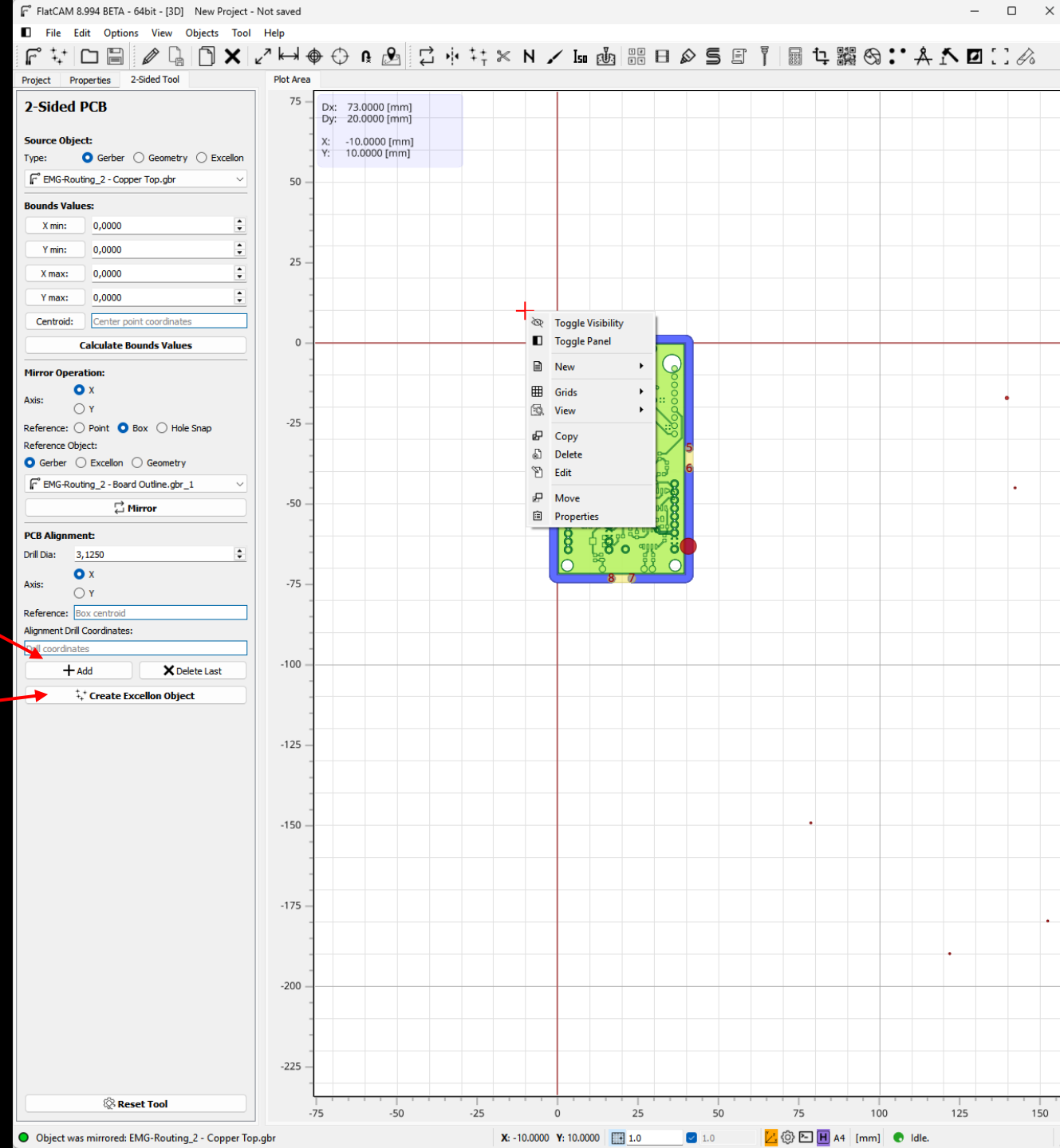
Select Drill dia: (usually 1 or 1.5mm)

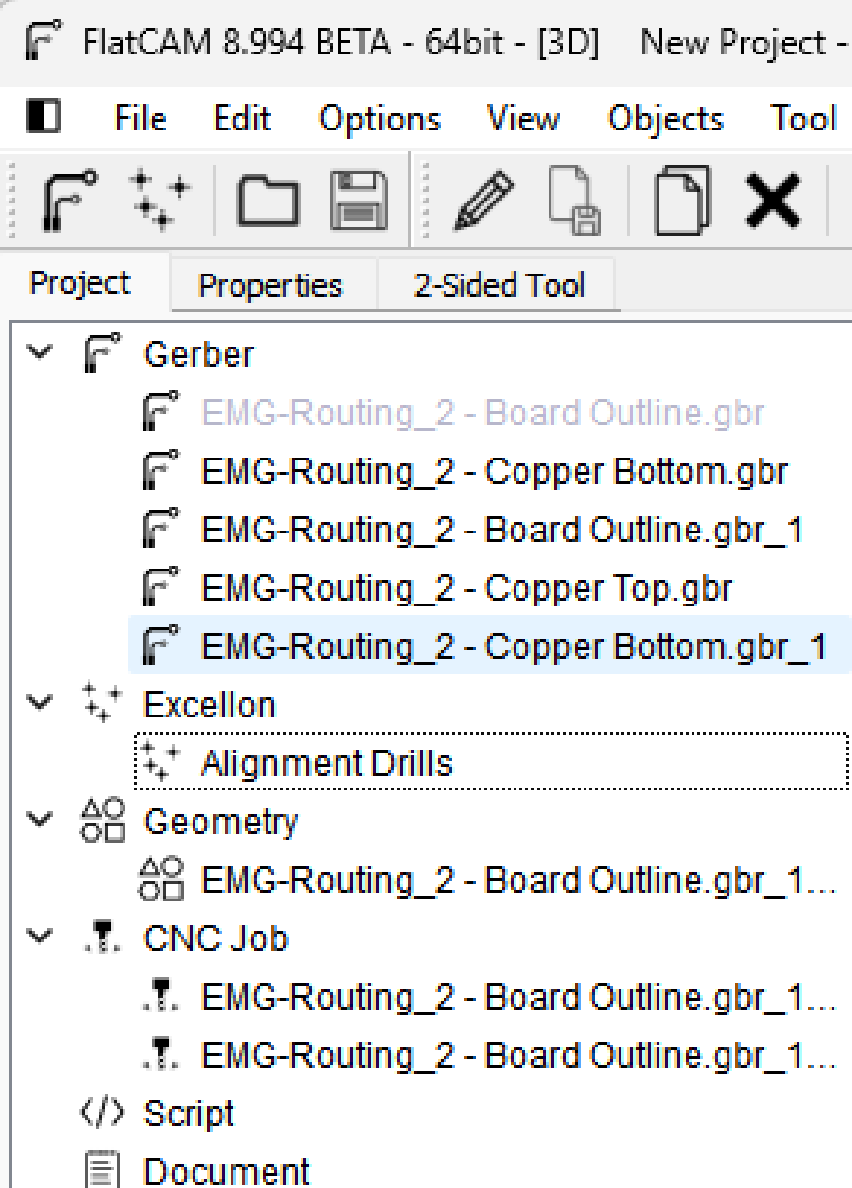


Select alignment drill coordinates by left clicking at the coordinate grid. Then press Add

Generate excellon object.

This coordinate will create 2 points on the copper plate for allignent on the wegstr when flipping the PCB





In project select the newly created gerber file that was mirrored and repeat step 1.2 Isolation routing

After finishing go back to project and select the Excellon file named "Alignment Drills" and repeat step 1.3 Drilling

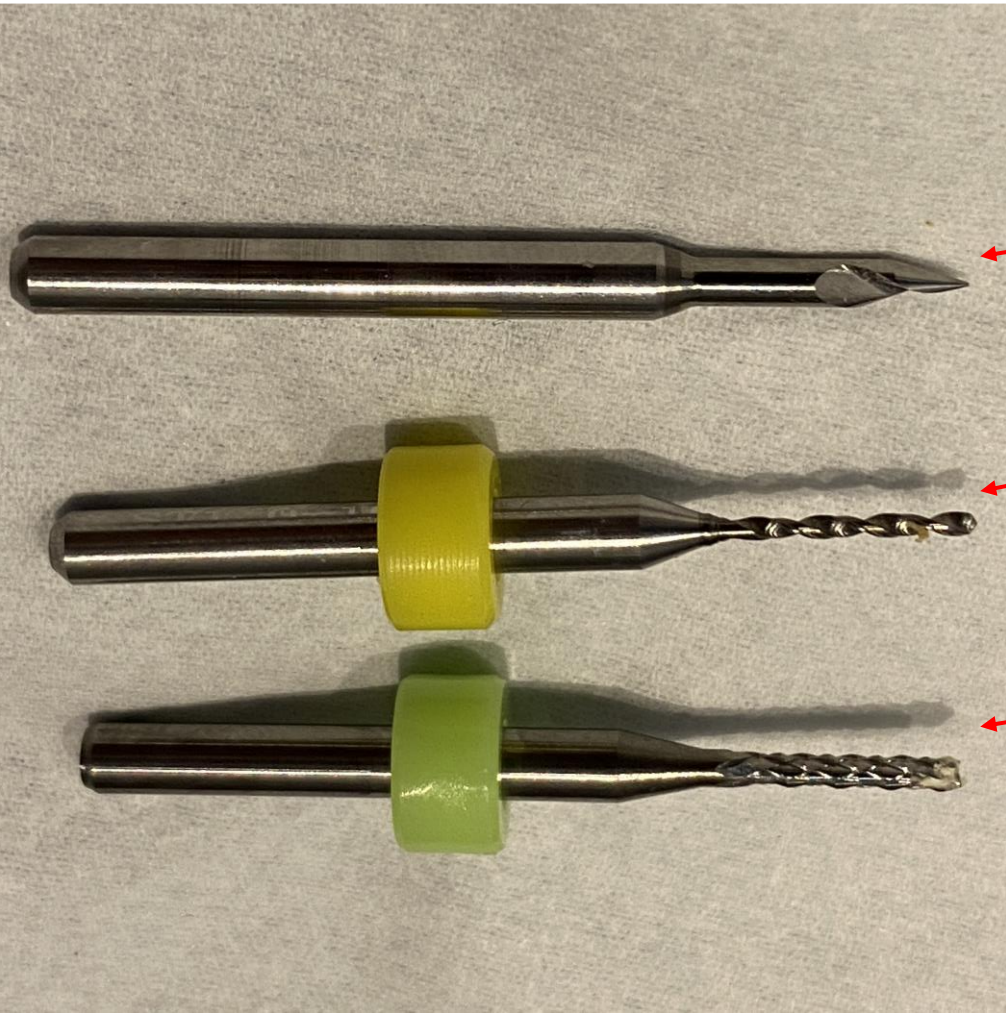
Visit the flatcam site for further information and learning more how everything works

flatcam: <http://flatcam.org/manual/index.html>

Wegster machine operation

Make sure to connect the USB-drive containing your Flatcam files knowing what each file contains makes this part easier to follow.

Different Bit types



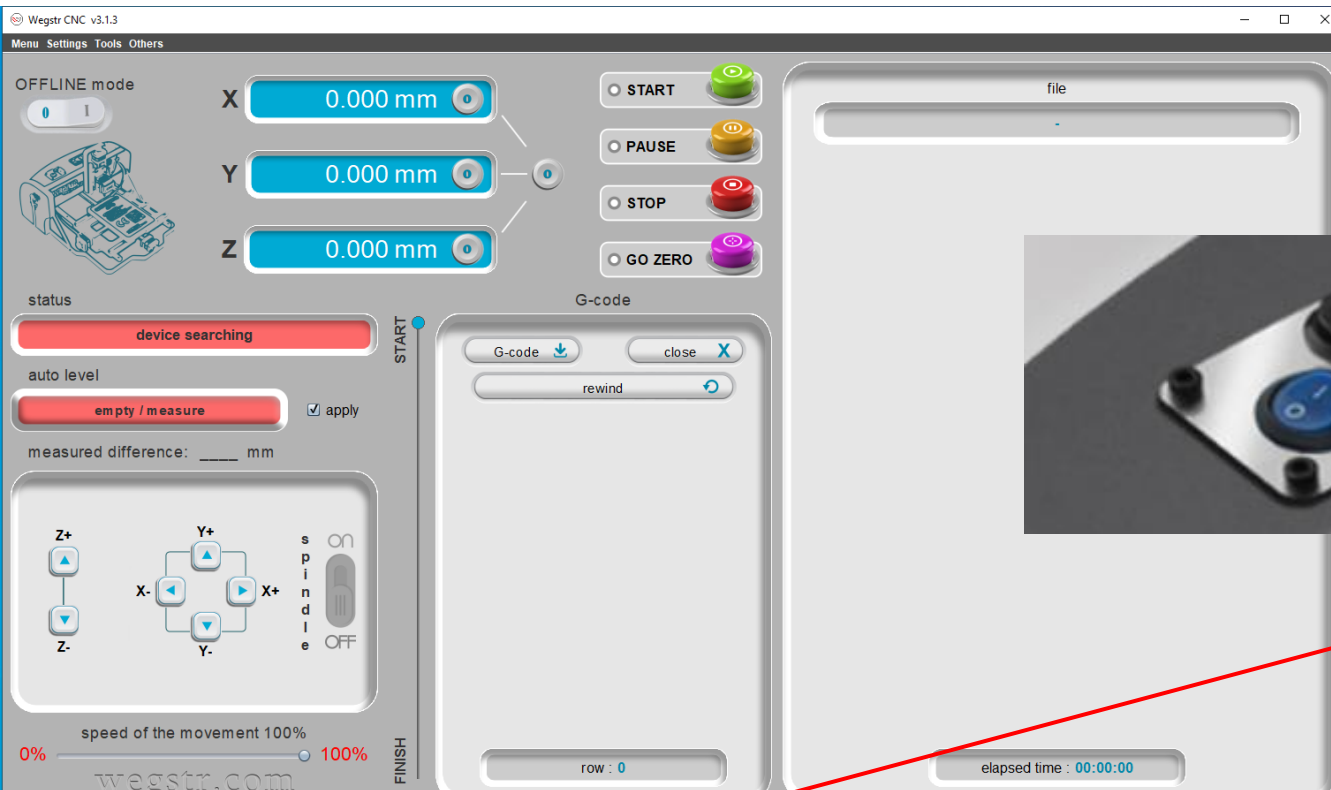
Spiral Engraving bit

Drill Bit

Flathead end mill bit

Bit uses

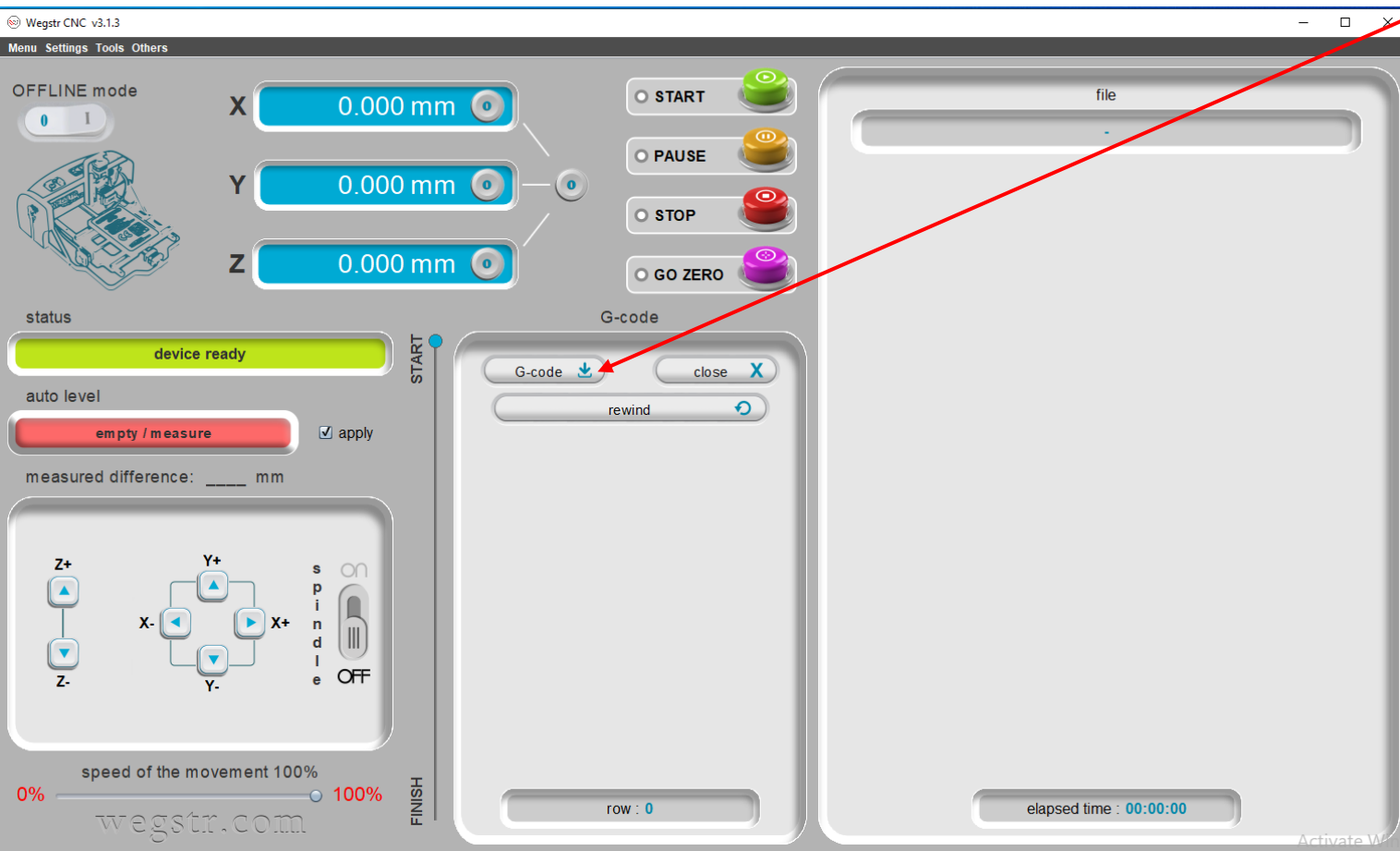
- Spiral engraving bit is used for isolation routing creating the traces on the pcb.
- Drill bit if u dont know what it does go home
- Flathead end mill bit used for clearing large copper areas and creating the outline of the PCB
- Files ending with .gbr uses the spiral engraving bit or flathead end mill bit and .drl will use the drill bit.



Connecting the software and hardware

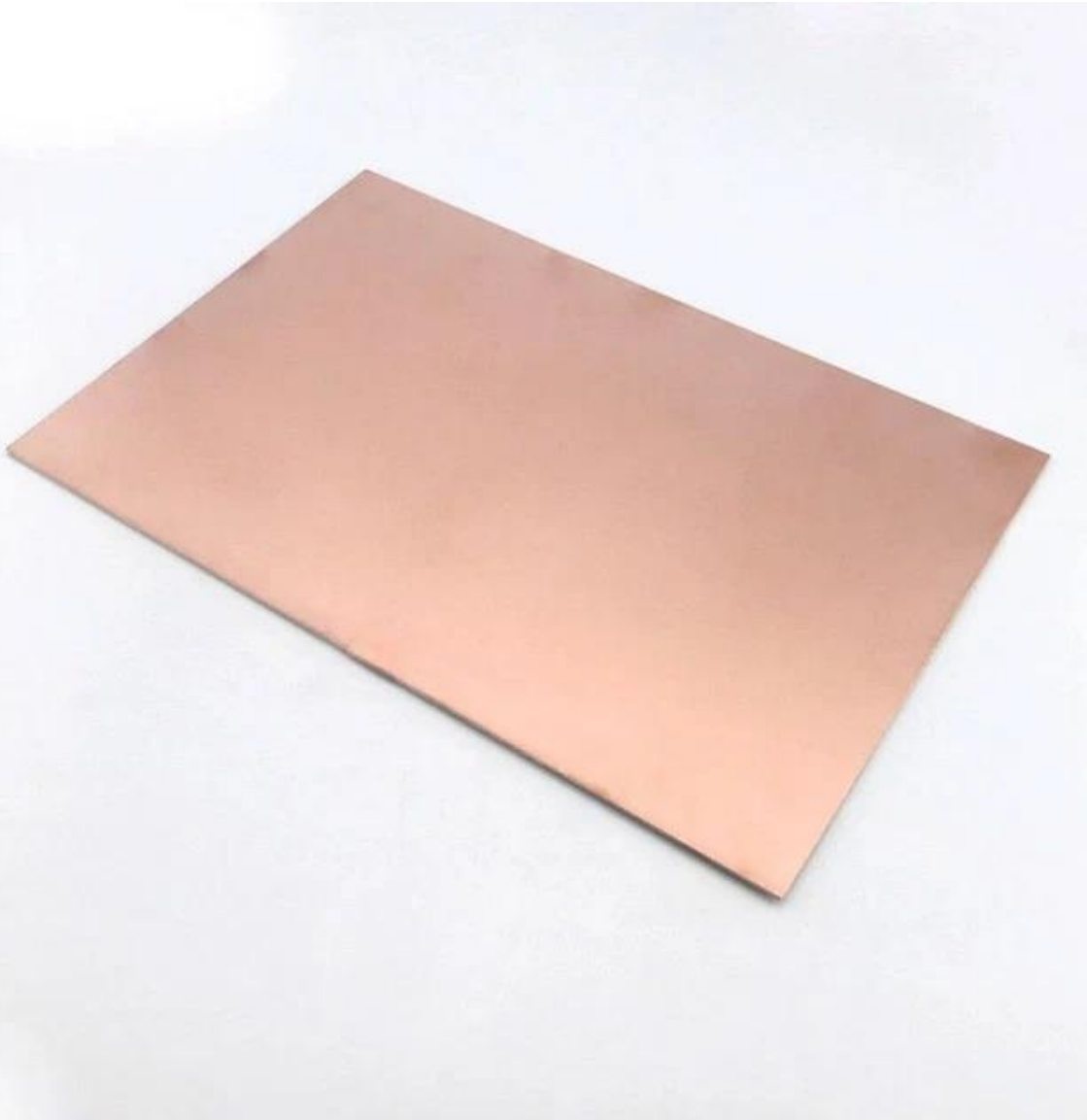
Turn on The wegstr status will become orange.

Press the start button and status will become green and read device ready



Select G-code open the Folder containing the flatcode files on the USB device. Make sure to do outline last. If creating a 2-sided PCB isolation routing should be made for both sides. Drill holes is only made once. Make sure to do outline last in both cases

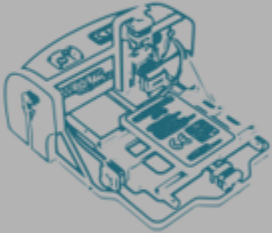
Origin for the Wegstr machine



Make sure to place the origin on a spot on the copper plate where the whole PCB is going to fit. It is good to know the size of the PCB that is being created.

OFFLINE mode

0 1



X

0.000 mm

Y

0.000 mm

Z

0.000 mm

START

PAUSE

STOP

GO ZERO

status

device ready

auto level

empty / measure

☒ apply

measured difference: ____ mm

START

G-code

G-code

close

rewind

Z+



Z-

X-

Y+



Y-

X+

spindle

on



OFF

speed of the movement 100%

0% 100%

wegstr.com

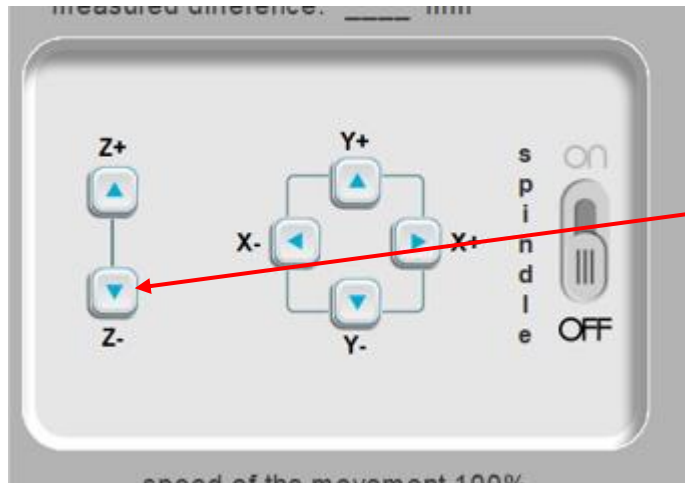
FINISH

row : 0

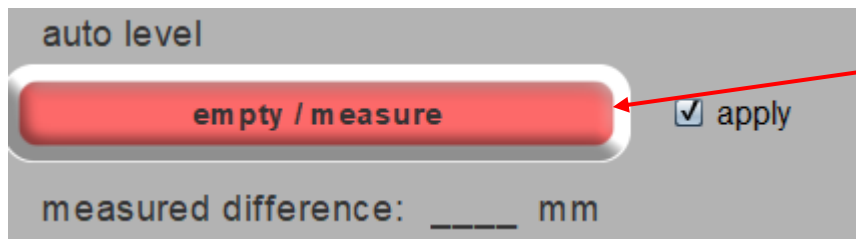
The Spindle can be moved around using the keyboard arrow buttons to move in the xy Direction or the xyz coordinates in the wegstr software. Z is used for moving the spindle Depth away or towards the copper plate and XY is for directional movement left right up down.



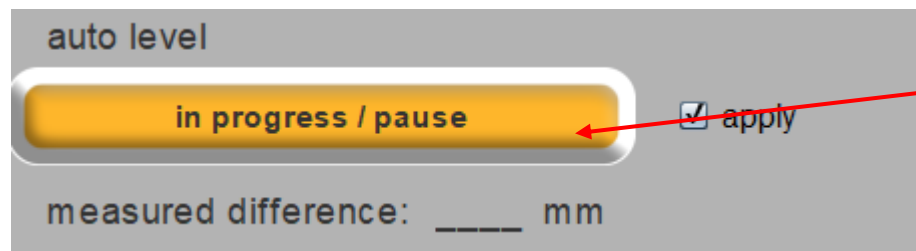
When the Spindle is on the origin spot of the users choosing then select the zero all button.



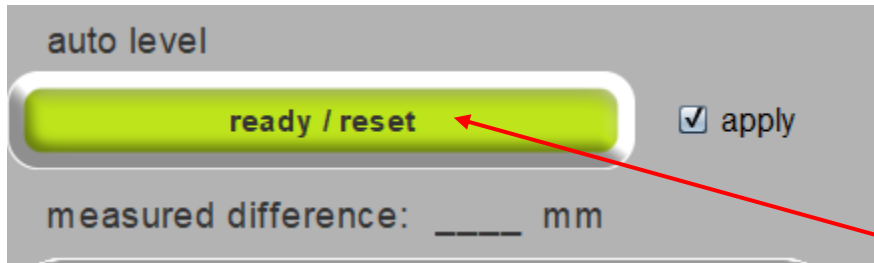
Move the spindle down to the copper using the Z buttons



Press empty/ measure to start auto leveling the spindle will start moving



It will change into this while auto leveling wait until it stops moving



When auto level is finished it will turn green and say ready/reset and stop moving



Press start to start the routing.

- When routing a file is finished move on to the next file by uploading it with g-code from the usb. Then zero the Z axis so it is touching the copper plate and start the next program again.



Only zeroes the Z-axis

Important Notes

- Makes sure that the spindel is only moving around the copper area.
- When zeroing the Z-Axis make sure that its moving towards copper and not the glass fiber under the copper surface.